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DSA0604 Slot B - MCQ

Assessment I - CO1

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1. A hospital dashboard displays patient age using point size and blood pressure using point color. Elderly patients appear larger, while blood pressure is represented using a blue-to-red gradient. Which visualization principle is primarily violated?

*

- ◯ A. Position should not encode quantitative values.
- ◉ B. Point size is unsuitable for ordered quantitative comparisons.
- ◯ C. Color gradients cannot represent quantitative values.
- ◯ D. Age should always be mapped to the x-axis.

2. A financial analyst uses different shapes to represent monthly stock prices over time. Users struggle to identify trends.
Which mapping would improve quantitative interpretation? *

- ◯ A. Map stock prices to shape.
- ◯ B. Map stock prices to color hue.
- ◉ C. Map stock prices to vertical position.
- ◯ D. Map stock prices to transparency.

3. A researcher maps customer satisfaction levels (Very Poor, Poor, Fair, Good, Excellent) to random colors without ordering.

The visualization becomes difficult to interpret because

*

- ◯ A. Satisfaction is nominal.
- ◉ B. Satisfaction is ordinal and requires ordered aesthetics.
- ◯ C. Random colors increase contrast.
- ◯ D. Color hue automatically implies ordering.

4. A climatologist maps rainfall intensity simultaneously to point size and color intensity.

The visualization becomes misleading because

*

- (selected) A. Redundant encoding may exaggerate perceived differences.
- B. Color cannot represent quantitative variables.
- C. Point size cannot encode quantitative data.
- D. Multiple aesthetics always improve interpretation.

5. An analyst represents employee departments using a continuous color gradient.

Which issue arises?

*

- (selected) A. Continuous scales imply ordering where none exists.
- B. Gradients improve categorical distinction.
- C. Departments are quantitative.
- D. Cartesian coordinates become invalid.

6. A scatter plot maps income to x-position and education level to y-position. Gender is represented using color.

Which aesthetic represents nominal data?

*

- A. x-position
- B. y-position
- (selected) C. Color

- ○ D. Position

7. A visualization maps annual sales to point transparency instead of position.

Why is this a poor choice?

*

- ○ A. Transparency has low perceptual accuracy.
- ○ B. Transparency cannot encode variables.
- ○ C. Transparency represents categories only.
- ◉ D. Position should only represent nominal data.

8. A chart uses texture patterns instead of colors to distinguish land-use categories.

The primary advantage is

*

- ○ A. Improved quantitative precision
- ◉ B. Better accessibility for color-blind users
- ○ C. Reduced memory usage
- ○ D. Faster rendering speedtion

9. A dataset contains Country, GDP, Population, and Continent.

Which variable should ideally be encoded using shape?

*

- ○ A. GDP
- ○ B. Population
- ◉ C. Continent

- D. GDP Growth

10.
Which visualization mapping produces the highest perceptual accuracy for comparing numerical values?
*

- A. Area
- B. Color satur
- (selected) C. Position along a common scale
- D. Transparency

11. A data scientist uses a logarithmic scale for household income.

Which reason best justifies this choice?

*

- A. Income contains many categories.
- (selected) B. Income spans several orders of magnitude.
- C. Log scales improve categorical comparis
- D. Logarithmic scales remove outliers.

12. A temperature visualization assigns identical colors to values between 20°C and 25°C.

The scale primarily reduces

*

- (selected) A. Precision
- B. Coordinate transformation
- C. Scatter density

D. Position accuracy

13. A continuous variable is mapped to five discrete colors.

Which problem is introduced?

*

A. False precision

B. Information loss

C. Axis inversion

D. Overplotting

14. A scatter plot uses identical symbol sizes for all observations while mapping sales to color intensity.

Which aesthetic is unused?

*

A. Position

B. Size

C. Color

D. Shape

15.
When transforming data values into visual positions, the mapping function is called
*

A. Coordinate system

B. Scale

C. Axis

D. Legend

16. A scientist rotates Cartesian coordinates by 45°.

Which property remains unchanged?

*

A. Relative distances

B. Scale labels

C. Color palette

D. Shape encoding

17. A visualization requires comparison of radial distances from a central point.

Which coordinate system is appropriate?

*

A. Cartesian

B. Polar

C. Parallel

D. Ternary

18. A business analyst compares quarterly revenues using pie slices arranged radially.

Which issue is most likely?

*

- (selected) A. Difficulty comparing angles accurately
- B. Cartesian distortion
- C. Incorrect logarithmic transformation
- D. Missing legends

19. A map uses latitude and longitude.

These coordinates are examples of

*

- A. Polar
- B. Cartesian
- (selected) C. Geographic coordinate system
- D. Logarithmic coordinates

20. Why are Cartesian coordinates preferred for scatter plots? *

- A. They improve color perception.
- (selected) B. They preserve independent quantitative dimensions.
- C. They require fewer data points.
- D. They eliminate outliers.

21. A microbiologist visualizes bacterial counts ranging from 100 to 10 million.

Which axis is most suitable?

*

- A. Linear
- B. Polar

- [x] C. Logarithmic
- [ ] D. Circular

22. Which statement about logarithmic axes is TRUE? *

- [ ] A. Equal distances represent equal differences.
- [ ] B. Equal distances represent equal ratios.
- [ ] C. Values cannot be negative.
- [x] D. Both B and C.

23. A student uses a log axis for exam scores between 60 and 100.

The visualization becomes

*

- [ ] A. More informative
- [x] B. Less meaningful because data span is small
- [ ] C. Easier to compare
- [ ] D. Better for categorical data

24. Applying nonlinear scaling mainly helps *

- [ ] A. Increase categorical distinction
- [x] B. Compress skewed numerical distributions
- [ ] C. Improve color accessibility
- [ ] D. Eliminate missing values

25. Which transformation best linearizes exponential growth? *

- ( ) A. Square root
- (•) B. Logarithm
- ( ) C. Reciprocal
- ( ) D. Mean normalization